Hayden Raymond Ramm
Born: 2001, Pasadena, California
Citizenship: United Kingdom / United States
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GitHub: https://github.com/haydenramm
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Education

2026– **University of Edinburgh**
PhD in Machine Learning
(Advisor: A. Storkey)

2024–2025 **University of Oxford**
Doctoral student in Atomic and Laser Physics

2020–2024 **University of Cambridge**
BA and MSci in Natural Sciences (Physics)

Publications / Talks / Posters

[1] **Ramm, H.**, Simon, P., Alexaki, P., Arran, C., Bingham, R., Goillot, A., Gudmundsson, J., Halliday, J., Lloyd, B., Los, E., Stergiou, V., Zhang, S., Gregori, G., & Charitonidis, N. (2026). An online data analysis framework for small-scale physics experiments. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, 1085, 171269.

[2] Ramm, H. Bayesian Statistics and Variational Inference, Lecture Notes (2026). *[Preprint]*

[3] Ramm, H. Beyond GPT: Teaching AI How to Think (2026). University of Oxford MPLS Ethics in AI Conference (Poster & Oral Presentation).

Industry

2026 **Observatory Sciences Ltd., Oxford** — Software Engineer
2025 **European Organisation of Nuclear Research (CERN), Geneva** — Software Engineer (Research)

Awards

2026 EPSRC DLA Scholarship (Department of Informatics, University of Edinburgh)
2024 ERC JETLAB Academic Scholarship (Department of Physics, University of Oxford)
2024 Trinity College Senior Scholarship (Trinity College, Cambridge)
2024 Trinity College Exam Prize (Trinity College, Cambridge)
2020 Gold Medallist, UK Chemistry Olympiad

Technical Proficiencies / Interests

- **[Experienced]** Python (PyTorch, Jax, TensorFlow/Keras, Pandas, Numpy), Bash / Linux & Kali Linux, uv
- **[Proficient]** C++, HTML / CSS
- **[Conversant]** JavaScript, C, OpenHands SDK
- Physics-informed machine learning, probabilistic machine learning / Bayesian neural networks, variational inference, continual learning, quantum field theory

Teaching

University of Edinburgh

2026– Department of Informatics & Department of Physics
- TA, *Probabilistic Machine Learning*
- TA, *Data Analysis and Machine Learning in Physics*

University of Oxford

2024–2026 Department of Physics (TA)